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(54) **TYRE WEAR MANAGEMENT FOR A VEHICLE**

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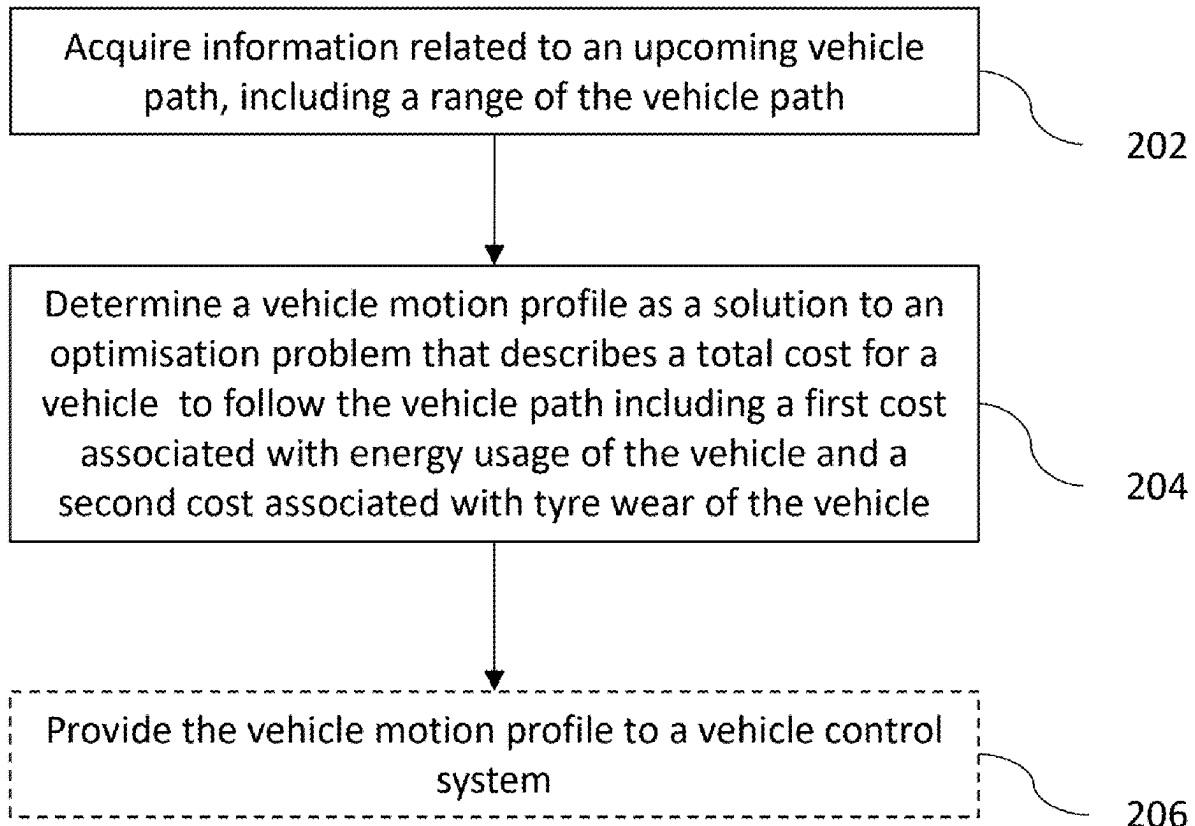
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(57)

ABSTRACT

A computer system determines a vehicle motion profile. The computer system has processing circuitry to acquire information related to an upcoming vehicle path, including a range of the vehicle path; and determine a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle to follow the vehicle path including a first cost associated with energy usage of the vehicle and a second cost associated with tyre wear of the vehicle. The total cost associated with the solution to the optimisation problem is below a threshold. The determined vehicle motion profile meets the range of the vehicle path.

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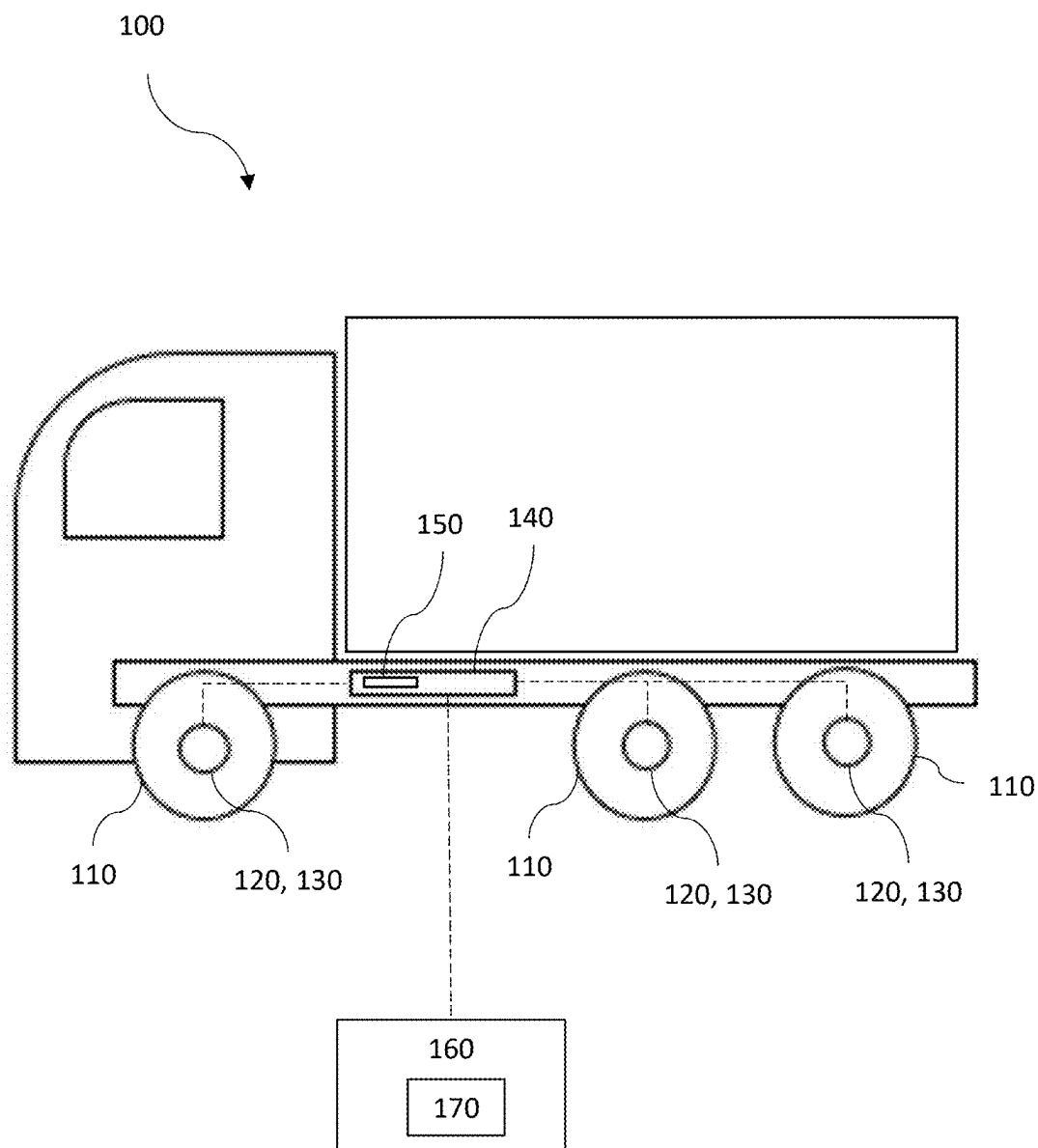


FIG. 1

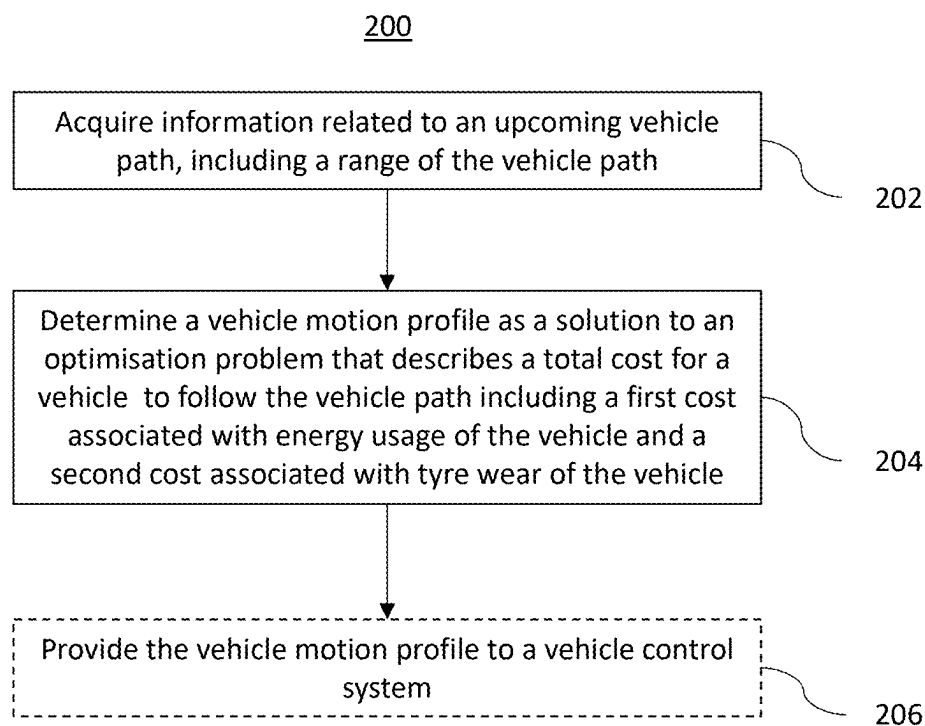


FIG. 2

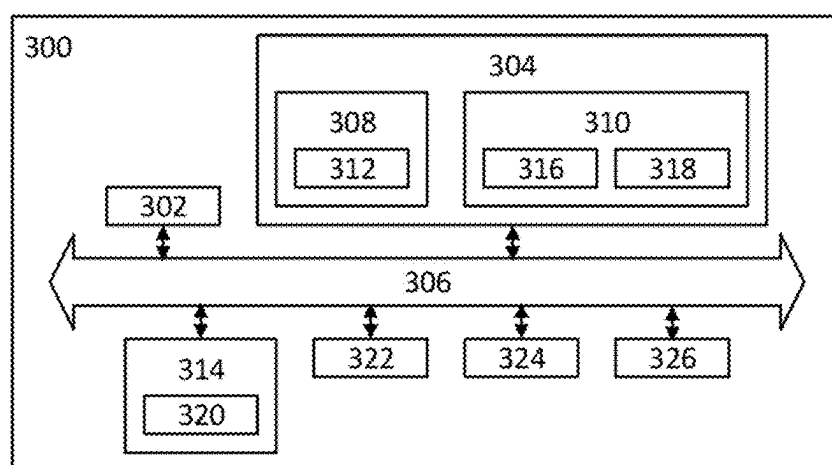


FIG. 3

TYRE WEAR MANAGEMENT FOR A VEHICLE

TECHNICAL FIELD

[0001] The disclosure relates generally to vehicle control. In particular aspects, the disclosure relates to tyre wear management for a vehicle. The disclosure can be applied in heavy-duty vehicles, such as trucks, buses, and construction equipment. In particular, the disclosure can be applied in multi-unit vehicle combinations with distributed propulsion and energy storage. Although the disclosure may be described with respect to a particular vehicle, the disclosure is not restricted to any particular vehicle.

BACKGROUND

[0002] Many vehicles today are run with vehicle motion profiles that seek to meet a certain goal without taking tyre wear into account. For example, predictive energy management, i.e., controlling vehicle motion by optimal torque (or power) distribution between different power sources and brakes considering upcoming road and traffic, can be intended to increase efficiency and range. This can include actions such as regenerative braking intended to increase energy efficiency.

[0003] However, certain strategies for increasing energy efficiency, such as regenerative braking, involve an increase in the axial torque on the wheels of the vehicle, leading to an increase in tyre wear. In this scenario, the benefits of energy efficiency will at some point be outweighed by the downsides of tyre wear.

[0004] It is therefore desired to develop a solution for vehicle motion management that addresses or at least mitigates some of these issues.

SUMMARY

[0005] This disclosure provides systems, methods and other approaches for determining a vehicle motion profile such that costs of different considerations, including tyre wear, are adequately balanced. In particular, information related to an upcoming vehicle path is acquired, including a range of the path. A solution to an optimisation problem, e.g. a non-linear optimal control problem (NOCP), is formulated relating to a total cost for the vehicle to follow the vehicle path, including a cost associated with energy usage of the vehicle and a cost associated with tyre wear of the vehicle. A solution to the optimisation problem is found such that the total cost is below a threshold and the range of the path is met. The solution is then implemented as a vehicle motion profile for the vehicle to follow the vehicle path, meaning that the costs of different considerations, including energy usage and tyre wear, are adequately balanced as the vehicle follows the path.

[0006] According to a first aspect of the disclosure, there is provided a computer system for determining a vehicle motion profile, the computer system comprising processing circuitry configured to acquire information related to an upcoming vehicle path, including a range of the vehicle path, and determine a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle to follow the vehicle path including a first cost associated with energy usage of the vehicle and a second cost associated with tyre wear of the vehicle, wherein the total cost associated with the solution to the optimisation

problem is below a threshold, and the determined vehicle motion profile meets the range of the vehicle path.

[0007] The first aspect of the disclosure may seek to provide a computer system that balances tyre wear with energy usage of a vehicle when planning a motion profile for a journey. The costs associated with energy usage of and tyre wear can be in contradiction, meaning that reducing one can result in an increase of the other, depending on the vehicle, control strategy and road. By determining a motion profile that ensures a total cost associated with energy usage and tyre wear is below a threshold, it can be ensured that operation of the vehicle when following the path does not exceed the cost. This means that different and potentially contradictory considerations, in particular energy usage and tyre wear, are adequately balanced during the journey.

[0008] Optionally in some examples, including in at least one preferred example, the vehicle path represents an entire journey for the vehicle, from a starting location to a destination location. A technical benefit may include that a motion profile can be determined for an entire journey, meaning energy usage and tyre wear can be adequately balanced beyond mere instantaneous or short-term manoeuvres usually handled by vehicle motion management systems. If these considerations are not made over an entire journey, the range of the vehicle may be reduced.

[0009] Optionally in some examples, including in at least one preferred example, the information related to the upcoming vehicle path comprises a friction profile, a curvature, and/or a vertical profile of the vehicle path. A technical benefit may include that a vehicle motion profile can be determined that takes various different parameters of the vehicle path into account, ensuring that both global and local factors across the route are considered when balancing energy usage and tyre wear.

[0010] Optionally in some examples, including in at least one preferred example, the processing circuitry is configured to implement the range of the vehicle path as a constraint of the optimisation problem. A technical benefit may include that a range requirement of a particular mission will always be met by the determined vehicle motion profile.

[0011] Optionally in some examples, including in at least one preferred example, the optimisation problem is a non-linear optimal control problem, NOCP. A technical benefit may include enhanced accuracy, robustness, and flexibility when performing the optimisation.

[0012] Optionally in some examples, including in at least one preferred example, the first cost represents CO₂ emissions and/or electrical energy usage. A technical benefit may include that environmental and efficiency factors of the vehicle are included in the optimisation such that a vehicle motion profile can be determined based on such considerations.

[0013] Optionally in some examples, including in at least one preferred example, the second cost represents particulate emissions from a tyre, friction loss from a tyre, and/or a cost associated with tyre maintenance. A technical benefit may include that environmental and performance factors related to the tyres of the vehicle are included in the optimisation such that a vehicle motion profile can be determined based on such considerations.

[0014] Optionally in some examples, including in at least one preferred example, the processing circuitry is configured to determine a cost associated with particulate emissions and/or friction loss from a tyre as a function of a longitudinal

slip associated with the one or more wheels. A technical benefit may include that the optimisation is performed with respect to a controllable parameter of the vehicle, meaning that the determined vehicle motion profile can be readily implemented in vehicle control.

[0015] Optionally in some examples, including in at least one preferred example, the vehicle motion profile comprises one or more of a speed of the vehicle, a trajectory of the vehicle, and a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle. A technical benefit may include that the vehicle can be controlled in numerous different ways to meet the determined vehicle motion profile and provide the desired balance between energy usage and tyre wear.

[0016] Optionally in some examples, including in at least one preferred example, the processing circuitry is further configured to provide the vehicle motion profile to a control system configured to determine one or more control inputs for the vehicle. A technical benefit may include that a control system of the vehicle is able to control the vehicle to provide the desired balance between energy usage and tyre wear.

[0017] Optionally in some examples, including in at least one preferred example, the one or more control inputs relate to one or more driven axles of the vehicle. A technical benefit may include that driven axles, where tyre wear caused by axial torque and/or longitudinal slip is especially higher, are controlled appropriately to meet the determined vehicle motion profile.

[0018] According to a second aspect of the disclosure, there is provided a vehicle comprising the computer system of any preceding example. The second aspect of the disclosure may seek to provide a vehicle capable of being operated to balance different and potentially contradictory considerations, in particular energy usage and tyre wear.

[0019] According to a third aspect of the disclosure, there is provided a computer-implemented method for determining a vehicle motion profile, the method comprising acquiring, by processing circuitry of a computer system, information related to an upcoming vehicle path, including a range of the vehicle path, and determining, by the processing circuitry, a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle to follow the vehicle path including a first cost associated with energy usage of the vehicle and a second cost associated with tyre wear of the vehicle, wherein the total cost associated with the solution to the optimisation problem is below a threshold, and the determined vehicle motion profile meets the range of the vehicle path.

[0020] The third aspect of the disclosure may seek to provide a computer-implemented method that balances tyre wear with energy usage of a vehicle when planning a motion profile for a journey. The costs associated with energy usage of and tyre wear can be in contradiction, meaning that reducing one can result in an increase of the other, depending on the vehicle, control strategy and road. By determining a motion profile that ensures a total cost associated with energy usage and tyre wear is below a threshold, it can be ensured that operation of the vehicle when following the path does not exceed the cost. This means that different and potentially contradictory considerations, in particular energy usage and tyre wear, are adequately balanced during the journey.

[0021] According to a fourth aspect of the disclosure, there is provided a computer program product comprising

program code for performing, when executed by processing circuitry, the computer-implemented method of the third aspect. The fourth aspect of the disclosure may seek to enable new vehicles and/or legacy vehicles to be conveniently configured, by software installation/update, to determine a vehicle motion profile that balances different and potentially contradictory considerations, in particular energy usage and tyre wear.

[0022] According to a fifth aspect of the disclosure, there is provided a non-transitory computer-readable storage medium comprising instructions, which when executed by processing circuitry, cause the processing circuitry to perform the computer-implemented method of the third aspect. The fifth aspect of the disclosure may seek to enable new vehicles and/or legacy vehicles to be conveniently configured, by software installation/update, to determine a vehicle motion profile that balances different and potentially contradictory considerations, in particular energy usage and tyre wear.

[0023] The disclosed aspects, examples (including any preferred examples), and/or accompanying claims may be suitably combined with each other as would be apparent to anyone of ordinary skill in the art. Additional features and advantages are disclosed in the following description, claims, and drawings, and in part will be readily apparent therefrom to those skilled in the art or recognized by practicing the disclosure as described herein.

[0024] There are also disclosed herein computer systems, control units, code modules, computer-implemented methods, computer readable media, and computer program products associated with the above discussed technical benefits.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] Examples are described in more detail below with reference to the appended drawings.

[0026] FIG. 1 schematically shows a side view of a vehicle according to an example.

[0027] FIG. 2 is a flow chart of a computer-implemented method according to an example.

[0028] FIG. 3 is a schematic diagram of a computer system for implementing examples disclosed herein.

[0029] Like reference numerals refer to like elements throughout the description.

DETAILED DESCRIPTION

[0030] The detailed description set forth below provides information and examples of the disclosed technology with sufficient detail to enable those skilled in the art to practice the disclosure.

[0031] Many vehicles today are run with vehicle motion profiles that seek to meet a certain goal without taking tyre wear into account. For example, predictive energy management, i.e., controlling vehicle motion by optimal torque (or power) distribution between different power sources and brakes considering upcoming road and traffic, is often intended to minimise energy consumption over an upcoming road horizon. This can include actions such as regenerative braking intended to increase energy efficiency. However, certain strategies for increasing energy efficiency, such as regenerative braking, involve an increase in the axial torque on the wheels of the vehicle, leading to an increase in tyre wear. In this scenario, the benefits of energy efficiency will at some point be outweighed by the downsides of tyre wear.

[0032] To remedy this, systems and methods are proposed for determining a vehicle motion profile. In particular, systems and methods are proposed in which information related to an upcoming vehicle path is acquired, including a range of the path. A solution is found to an optimisation problem that describes a total cost for a vehicle to follow the path, including a cost associated with energy usage of the vehicle and a cost associated with tyre wear of the vehicle. The solution is found such that the total cost associated with the solution is below a threshold and the range of the path is met. The solution is a vehicle motion profile that can be implemented to control a vehicle, and ensures that the costs of different considerations, including energy usage and tyre wear, are adequately balanced.

[0033] FIG. 1 schematically shows a side view of an example vehicle 100 of the type considered in this disclosure. The vehicle 100 may be any suitable form of vehicle. For example, the disclosure can be applied in heavy-duty vehicles, such as trucks, buses, construction equipment, and multi-unit vehicle combinations, in personal vehicles such as cars, vans, or motorbikes, or in any other suitable form of vehicle. The vehicle 100 comprises a number of axles, each generally having two or more wheels 110. Whilst three axles are shown, it will be appreciated that any suitable number of axles may be provided. It will also be appreciated that any number of the axles may be driven axles.

[0034] The vehicle 100 may comprise one or more sources or propulsion. For example, the vehicle 100 may comprise one or more electrical machines 120 such as electric motors and/or generators. The vehicle 100 may comprise one or more batteries (not shown) configured to provide power to the electrical machines 120. In some examples, the vehicle 100 may also include another source of propulsion, for example an internal combustion engine (ICE). The vehicle 100 also comprises a drivetrain (not shown) to deliver mechanical power from the propulsion source (the electrical machines 120 or the ICE) to the wheels 110.

[0035] The electrical machines 120 are configured to drive, e.g. provide torque and/or steering to, one or more axles or individual wheels 110 of the vehicle 100. The electrical machines 120 can supply either a positive (propulsion) or negative (braking) force. The use of electrical machines 120 to supply a negative force is known as regenerative braking, in which case the electrical machines 120 may be operated as generators, in order to recover energy during braking.

[0036] Furthermore, the vehicle 100 may comprise one or more sets of service brakes 130. The service brakes 130 can supply a negative (braking) force. The service brakes 130 may be, for example, frictional brakes such as pneumatic brakes. Pneumatic brakes use a compressor to fill the brake with air, which may be powered by the batteries. In some examples, the brakes may be electro-mechanical brakes. The energy recovered from regenerative braking by the electrical machines 120 can be stored in the batteries, and so regenerative braking may generally be preferred over using service brakes.

[0037] In some examples, the vehicle 100 may be a vehicle combination comprising a number of units, including a tractor unit and at least one trailing unit. A tractor unit is generally the foremost unit in a vehicle combination, and may comprise the cabin for the driver, including steering controls, dashboard displays and the like. Generally, the tractor unit is used to provide propulsion power for the

vehicle combination 100. A trailing unit is generally used to store goods that are being transported by the vehicle combination 100. A trailing unit may be a truck, trailer, dolly and the like. A trailing unit may also provide propulsion to the vehicle combination 100. In such examples, each unit may comprise its own electrical machines 120, batteries, service brakes 130, and the like. In this way, all units may provide propulsion to the vehicle combination 100.

[0038] When the vehicle 100 is in motion, the wheels (or indeed tyres) 110 of the vehicle 100 experience slip. Slip can be expressed as the body slip on the vehicle 100 as a whole, or the slip on a given wheel 110, which can be divided into longitudinal and lateral slip. These parameters are known in the art, and not discussed in detail here. However, it is noted that the longitudinal slip s_{xik} and lateral slip s_{yik} on a wheel 110 may be given by:

$$s_{xik} = \frac{R_{ik}\omega_{ik} - v_{wxik}}{|R_{ik}\omega_{ik}|} \quad (1)$$

$$s_{yik} = \frac{v_{wyik}}{|R_{ik}\omega_{ik}|} \quad (2)$$

where R is the tyre radius, w is the wheel rotational speed, v_{wxik} and v_{wyik} are the wheel hub velocity components in the wheel local coordinate system, k is the wheel index, and i is the unit index in the case of a multi-unit vehicle combination.

[0039] Returning to FIG. 1, the vehicle 100 includes a controller 140 comprising processing circuitry 150. The controller 140 is configured to control components of the vehicle, for example the electrical machines 120. FIG. 1 shows a common controller 140 for all electrical machines 120 of the vehicle 100, however it will be appreciated that each electrical machine 120 may have its own respective controller 140. In many cases, the controller 140 may be implemented in the structure of the electrical machine 120 itself. The controller 140 may be a microcontroller. In examples where the vehicle 100 is a vehicle combination, the vehicle 100 may include a global controller and a plurality of unit controllers, for example a controller for each unit. Vehicle motion management may therefore be available on a unit level to receive requests from a manual or virtual driver to coordinate the propulsion, braking and steering.

[0040] The controller 140 may receive control signals from a computer system 160 comprising processing circuitry 170. The computer system 160 may be a vehicle control unit configured to perform various vehicle (unit) control functions, such as vehicle motion management. The computer system 160 may be local to the vehicle 100, or may be a remote system, implemented at a distance from the vehicle 100. The computer system 160 may be communicatively coupled to the controller 140 in any suitable way, for example via a circuit or any other wired, wireless, or network connection known in the art. Furthermore, the communicative coupling may be implemented as a direct connection between the controller 140 and the computer system 160, or may be implemented as a connection via one or more intermediate entities.

[0041] One function of the controller 140 and the computer system 160 is to provide control inputs for the vehicle 100, for example torque, force, or slip requests. These control inputs should enable a requested manoeuvre for the

vehicle 100, for example, straight-line driving, cornering, braking and the like, whilst ensuring safe and efficient motion of the vehicle 100. In many cases, these control inputs are determined to meet a certain goal, for example energy efficiency, safety, range, etc. However, tyre wear is not always taken into account.

[0042] When a vehicle is in operation, the interaction between the tyres and the surface on which the vehicle is driven causes the tyres to break down into small particles. This is particularly the case in acceleration, braking, and turning manoeuvres. This has several negative impacts. One impact is that the broken down particles are emitted into the environment, polluting air, water, and soil. A typical tyre is made of a blend of natural rubber, synthetic rubber, plastic polymers, metals, and other chemical compounds. Particulate emissions of these materials can be harmful to the environment. Another impact is that, as tyre wear increases, the friction provided by the tyre is reduced. This is because the tread depth of the tyre is reduced as the tyre is worn down. As such, a tyre is not able to perform to its full capacity as tyre wear increases. Another impact is that, when a tyre becomes worn beyond its useful performance, it must be maintained and/or replaced. This causes a further increase in the environmental impact of tyre wear, as more tyres must be produced and transported and the vehicle must travel to a maintenance location.

[0043] These negative impacts are not often taken into account when determining a vehicle motion profile for a vehicle 100. A vehicle motion profile includes instructions and parameters for a vehicle 100 to complete a given journey, based on information about the journey such as a starting location, a destination location, a total distance, a friction profile, a curvature, and/or a vertical profile of the path to be travelled. A vehicle motion profile may include a speed of the vehicle 100, a trajectory of the vehicle 100, and a power split between propulsion and braking systems and subsystems of the vehicle 100. The motion of a vehicle 100 together with the use of the relevant components (e.g. electrical machines 120, batteries, service brakes 130, and the like) can be planned to meet a certain goal.

[0044] One approach to determining a vehicle motion profile is predictive energy management, i.e., controlling vehicle motion by optimal torque (or power) distribution between different power sources and brakes considering upcoming road and traffic in order to minimise energy consumption over an upcoming road horizon. This can include actions such as regenerative braking intended to increase energy efficiency. This can be achieved, for example, using brake-blending strategies of BEVs, where the entire brake request may be met by different driven axles to increase range and reduce energy consumption of BEVs. However, certain strategies for increasing energy efficiency, such as regenerative braking, involve an increase in the axial torque on the wheels 110 of the vehicle 100, leading to an increase in tyre wear. In this scenario, the benefits of energy efficiency will at some point be outweighed by the downsides of tyre wear.

[0045] To remedy this, systems and methods are proposed for determining a vehicle motion profile in which both energy usage and tyre wear are taken into account. In particular, a solution is found to an optimisation problem that describes a total cost for a vehicle to follow an upcoming vehicle path including a cost associated with energy usage of the vehicle and a cost associated with tyre wear of

the vehicle. The optimisation problem may be an optimal control problem (OCP). The OCP may be a non-linear optimal control problem (NOCP), although it is also possible to formulate a linear OCP that provides a solution of sufficient accuracy.

[0046] The solution is found such that the total cost associated with the solution is below a threshold and the range of the path is met. The solution is a vehicle motion profile that can be implemented to control a vehicle 100, and ensures that the costs of different considerations, including energy usage and tyre wear, are adequately balanced. According to some examples, the vehicle motion profile comprises a target tyre force trajectory and/or a target wheel slip trajectory for at least two wheels to be followed along the upcoming vehicle path. The vehicle motion profile may comprise a target vehicle unit acceleration and/or a target vehicle speed to be followed along the upcoming vehicle path. Both the target force trajectory and the acceleration profile may be generated under slip constraints, which means that the vehicle can be controlled to behave according to the target force trajectory and/or the acceleration profile without experiencing wheel slip in excess of the configured slip limits. The vehicle motion profile may comprise a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle. The solution minimises the total cost of energy usage and tyre wear of the vehicle 100 along the path.

[0047] To this end, a first cost in the optimisation problem is associated with energy usage of the vehicle 100, and a second cost in the optimisation problem is associated with tyre wear of the vehicle 100. The first cost may be formulated to represent CO₂ emissions and/or electrical energy usage. The second cost may be formulated to represent particulate emissions from a tyre, tyre tread depth reduction, and/or a cost associated with tyre maintenance. The first and second costs may be formulated in any suitable manner, for example based on modelling, historical data or experiments.

[0048] In some examples, the first cost associated with energy usage can be determined based on CO₂ emissions. In some examples, this can be expressed as a function of the fuel equivalent force F_f based on a value that converts fuel energy to CO₂. This value may be constant C_{diesel} that expresses a cost in terms of the CO₂ content of the fuel per unit energy content of the fuel. For example, for a value of CO₂ content of the fuel per litre of 2.68 kg/l, and a value of energy content of the fuel energy content of the fuel of 38.6e6 MJ/l, a value of C_{diesel} will be 6.9×10^{-8} kg/J. It will be appreciated that a cost associated with CO₂ emissions can be determined in any suitable manner.

[0049] In some examples, the first cost associated with energy usage can be determined based on electrical energy usage. In some examples, this can be expressed as a function of the friction service brake force F_{br} , and/or the dissipation force of the electric driveline F_{del} based on a value C_{el} that expresses a cost per unit of electrical energy. It will be appreciated that a cost associated with electrical energy usage can be determined in any suitable manner.

[0050] In some examples, the second cost associated with a tyre wear can be determined based on modelling particulate emissions and/or tyre tread depth reduction. It is noted that a number of different factors may influence tyre wear, such as axle load, temperature, tyre pressure, lateral slip, longitudinal slip, longitudinal slip rate, sliding distance, tyre material, tyre tread shape, tyre size, road type, and vehicle

velocity speed. However, many of these factors are not controllable during driving of the vehicle. Therefore, for the purposes of the present disclosure, a controllable parameter is considered. In particular, the control parameter may be an axial torque (positive or negative) and/or a longitudinal slip associated with the one or more wheels **110** of the vehicle **100**.

[0051] Particulate emissions E_{wear} from a tyre, expressed in kilograms, may be determined as follows:

$$E_{wear} = \frac{W_{tyre} K_{wear} d_s s_{xik}^2}{1000} \quad (3)$$

where W_{tyre} is the tyre width, K_{wear} is a fitting factor, and d_s is a distance increment (e.g. road distance step size of the recorded data). The longitudinal slip s_{xik} can be determined in any suitable manner, for example using equation (1), in the manner described in PCT patent application no. WO 2022/106004 in the name of Volvo Truck Corporation, based on a nonlinear tyre model, or linearly according to “*An Empirical Tire-Wear Model for Heavy-Goods Vehicles*”, June 2021, Tire Science and Technology 50(3).

[0052] Alternatively, particulate emissions E_{wear} for an axle, expressed in kilograms, may be determined based on a linear tyre model as follows:

$$E_{wear, axle} = n_{tyres} \frac{W_{tyre} K_{wear} d_s \left(\frac{F_{x, tyre}}{C_{s, tyre}} \right)^2}{1000} \quad (4)$$

where $F_{x, tyre}$ is the longitudinal force on the tyre, n_{tyres} is the number of tyres on the axle, and $C_{s, tyre}$ is a characteristic given by:

$$C_{s, tyre} = 10 F_{z, tyre} \quad (5)$$

where $F_{z, tyre}$ is the vertical force on the tyre.

[0053] A corresponding tyre tread depth reduction d_h may be determined as follows:

$$d_h = \frac{E_{wear}}{2 R W_{tyre} \pi D_{tyre}} \quad (6)$$

where D_{tyre} is the tyre material density.

[0054] Based on these values, a cost associated with tyre wear can be formulated accordingly. For example, a cost of tyre wear can be formulated based on the tyre tread depth reduction d_h and the tyre tread depth at the beginning of tyre life. A cost rate of tyre wear $C_{tireWearAxle, l}$ of axle l can be formulated based on the cost of tyre wear and the distance travelled. Moreover, the term of the tyre wear cost is a convex function of tyre torque, longitudinal force, or longitudinal slip, therefore, it does not cause additional computational load on existing predictive energy management strategies.

[0055] When determining the technical costs discussed herein, they may be normalised such that they can be compared in a relevant manner. This may be achieved in any

suitable manner. In one example, the technical costs may be normalised based on a unit of currency to express them financially. For example, weighting factors in the costs may be set such that the energy efficiency and tyre wear have the same unit which is cost-rate (of time or of distance, Euro/s or Euro/m). In this case, the tyre wear cost is a function of the vehicle type, axle loads, tyre type and size, price of tyre, services, and maintenance, as well as physics-based validated models to relate vehicle usage to the specific axle tyre wear via tyre longitudinal slip.

[0056] It will be appreciated that the costs discussed above are examples only, and an appropriate cost to describe the effects of axial torque and longitudinal slip may be formulated by an operator in any suitable manner.

[0057] Let s be a vehicle path variable (e.g. the distance travelled along the path). An NOCP can then be defined to determine, as a function of the distance travelled along the path, one or more of an engine force F_e provided by an ICE, a wheel force F_{mwk} provided by the k^{th} electrical machine **120**, a friction force F_{br} provided by the service brakes, an integer gear γ_e of the ICE, an integer gear γ_m of the electrical machines **120**. An NOCP can be defined, e.g., as follows: find $F_e(s)$, $F_{mwk}(s)$, $F_{br}(s)$, $\gamma_e(s)$, $\gamma_m(s)$, $k=1 \dots n_{em}$ to minimise:

(7)

$$J^{nl} = \int_{s_0}^{s_f} \left(F_f(\cdot) C_{diesel} + (F_{br}(s) + F_{del}(s)) C_{el} + \sum_{l=1}^{l=n_{dAxles}} C_{tireWearAxle, l} \right) ds$$

subject to:

[0058] Fuel equivalent force constraints:

(8)

$$F_f(F_e(s), v_x(s), \gamma_e(s), s) = \frac{1}{v_x(s)} \sum_{i=0}^5 \sum_{j=0}^5 a_{ij} \left(\frac{r_e(\gamma_e(s))}{R_w} v_x(s) \right)^i \left(\frac{R_w}{r_e(\gamma_e(s))} F_e(s) \right)^j$$

[0059] Vehicle longitudinal dynamics in space domain (road grade $a(s)$ positive downhill) including the effect of road curvature:

(9)

$$\frac{dv_x(s)}{ds} = \frac{1}{mv_x(s)} \left(-F_{br}(s) + \eta_{te} F_e(s) + F_{mw}(s) + mg \sin \alpha(s) - mg f_r \cos \alpha(s) - 0.5 \rho_a A_f c_d v_x(s)^2 \right)$$

[0060] Battery state of charge (SOC) dynamics:

$$\frac{dsoc(s)}{ds} = -\frac{1}{E_{bmax}} \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + \frac{R v_x(s)}{V_b^2} \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + F_a(s) \right)^2 \right) + F_a \quad (10)$$

[0061] Time dynamics:

(11)

$$\frac{dt(s)}{ds} = \frac{1}{v_x(s)}$$

[0062] Initial states:

$$v(s_0) = v_0, soc(s_0) = soc_0, t(s_0) = t_0 \quad (12)$$

[0063] Electrical machine (k=1 . . . n_{em}) consumption using two 5th degree fitting functions for positive and negative torques:

$$F_{mck}(s) = \frac{1}{v_x(s)} \sum_{i=0}^5 \sum_{j=0}^5 h_{ijk}^+ \left(\frac{r_{mk}(\gamma_{mk}(s))}{R_w} v_x(s) \right)^i \left(\frac{R_w F_{mvk}(s)}{r_{mk}(\gamma_{mk}(s)) \eta_{tmk}} \right)^j, \quad (13)$$

$$F_{mvk}(s) > 0,$$

$$F_{mck}(s) = \frac{1}{v_x(s)} \sum_{i=0}^5 \sum_{j=0}^5 h_{ijk}^- \left(\frac{r_{mk}(\gamma_{mk}(s))}{R_w} v_x(s) \right)^i \left(\frac{R_w F_{mvk}(s) \eta_{tmk}}{r_{mk}(\gamma_{mk}(s))} \right)^j, \quad (14)$$

$$F_{mvk}(s) \leq 0$$

[0064] Dissipation equivalent force of the electric driveline:

$$F_{del}(s) = - \sum_{k=1}^{k=n_{em}} F_{mvk}(s) + \sum_{k=1}^{k=n_{em}} F_{mck}(s) + \frac{R v_x(s)}{V_b^2} \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + F_a(s) \right)^2 \quad (15)$$

[0065] ICE torque upper limit:

$$\frac{R_w F_e(s)}{r_e(\gamma_e(s))} - \min \left\{ \sum_{j=0}^3 \left(\frac{r_e(\gamma_e(s))}{R_w} v_x(s) \right)^j b_{ij}^e, i = 1, \dots, 4 \right\} \leq 0 \quad (16)$$

[0066] Electrical machine torque upper and lower limits (k=1 . . . n_{em}):

$$\frac{R_w F_{mvk}(s)}{r_{mk}(\gamma_{mk}(s)) \eta_{tmk}} - \min \left\{ \sum_{j=0}^3 \left(\frac{r_{mk}(\gamma_{mk}(s))}{R_w} v_x(s) \right)^j b_{ijk}^m, i = 1, 2 \right\} \leq 0 \quad (17)$$

$$- \frac{R_w F_{mvk}(s) \eta_{tmk}}{r_{mk}(\gamma_{mk}(s))} + \max \left\{ \sum_{j=0}^3 \left(\frac{r_{mk}(\gamma_{mk}(s))}{R_w} v_x(s) \right)^j b_{ijk}^m, i = 3, 4 \right\} \leq 0 \quad (18)$$

[0067] Arrival time:

$$t(s_f) - t_{ref}(s_f) \leq 0 \quad (19)$$

[0068] Battery SOC limits:

$$soc_{min} - soc(s) \leq 0 \quad (20)$$

$$soc(s) - soc_{max} \leq 0 \quad (21)$$

[0069] Battery power limits:

$$p_{bmin} - v_x(s) \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + \frac{R v_x(s)}{V_b^2} \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + F_a(s) \right)^2 + F_a(s) \right) \leq 0$$

$$0 soc_{min} - soc(s) \leq 0 \quad (23)$$

$$v_x(s) \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + \frac{R v_x(s)}{V_b^2} \left(\sum_{k=1}^{k=n_{em}} F_{mck}(s) + F_a(s) \right)^2 + F_a(s) \right) - p_{bmax} \leq 0$$

[0070] Speed limits:

$$v_{min}(s) - v_x(s) \leq 0 \quad (24)$$

$$v_x(s) - v_{max}(s) \leq 0 \quad (25)$$

[0071] ICE torque or force lower limit:

$$-F_e(s) \leq 0 \quad (26)$$

[0072] Friction brake lower limit:

$$-F_{br}(s) \leq 0 \quad (27)$$

[0073] The notations in the table below apply to the above equations:

A_f	Front area of the vehicle	R	Batteries electric resistance
a_{ij}	Coefficients of fitted surface	r_e	Total gear ratio of the gearbox and differential related to ICE
b_{ij}^e	Fitting curve coefficients of the maximum ICE torque	r_m	Total gear ratio of the gearbox and differential related to EM
b_{ij}^m	Fitting curve coefficients of the maximum and minimum EM torques	R_w	Wheel effective radius
c_d	Drag coefficient	s	Distance travelled
E_{bmax}	Max. battery energy capacity	s_f	Final distance (end of the trip)
F_a	Equivalent force of the vehicle auxiliaries	soc	Batteries' state of charge
F_{br}	Friction service brake force	s_0	Initial distance
F_{del}	Dissipation force of electric driveline	t	Time

-continued

F_e	Engine force	t_{ref}	Reference trip time
F_f	Fuel equivalent force	u	Decision variables or the input vector of the optimal control problem
F_{mck}	Input equivalent force to k^{th} EM ($v_x F_{mc}$ yields the power consumed by EM)	V_b	Batteries voltage
F_{mwk}	Wheel force coming from k^{th} EM	v_x	Longitudinal velocity
f_r	Rolling resistance coefficient	x	State vector
F_{xwij}	Wheel longitudinal force of i^{th} unit and j^{th} axle	α	Road grade positive downhill
F_{ywij}	Wheel lateral force of i^{th} unit and j^{th} axle	γ_e	Gear number of ICE
F_{zij}	Axle vertical force of i^{th} unit and j^{th} axle	γ_m	Gear number of EM
g	Gravitational acceleration	η_{te}	Efficiency of ICE transmission
h_{ij}^+	Fitting function coefficients of the positive EM power consumption	η_{tm}	Efficiency of EM transmission
h_{ij}^-	Fitting function coefficients of the negative EM power consumption	μ	Road friction coefficient
J^{nl}	Nonlinear cost function	ρ_a	Air density
m	Gross combination mass		
P_{bmax}	Batteries maximum power		
P_{bmin}	Batteries minimum power		

[0074] Equation (7) is the cost function, where n_{dAxes} is the number of driven axles. Equations (7) to (27) represent a NOCP with inputs u which are the same as the decision variables and states:

$$x = [v_x, soc, t] \quad (28)$$

where v_x is the longitudinal speed of the tractor at its center of gravity, soc is state of charge and t is time. The problem may be solved in real time by sequential programming.

[0075] If the vehicle is a BEV and the distance to the destination is less than the calculated nominal range then mission is not range-critical, therefore, the nonlinear optimal control problem of predictive cost management (NOCP PCM) with a tyre wear cost term included in the cost function can be directly used as the preferable controller. Moreover, if the mission is range-critical, predictive cost management (PCM) can still be used since it includes a constraint to meet the feasible requested range of the mission (equations 20 and 21). For a hybrid vehicle, it may be preferable to use the NOCP PCM as the controller.

[0076] It will be appreciated that the constraints can be formulated in different ways mathematically with similar technical effects. It is furthermore appreciated that the type of NOCP described above can then be solved by computer-based solution methods known in the art.

[0077] FIG. 2 is a flow chart of a computer-implemented method 200 according to an example. The method 200 is for determining a vehicle motion profile for a vehicle, such as the vehicle 100, to complete a journey. The method 200 enables costs associated with tyre wear and other factors, including energy usage, to be balanced appropriately. The method 200 may be implemented by processing circuitry of a computer system (e.g., the processing circuitry 150 of the controller 140, or the processing circuitry 170 of the computer system 160 described in relation to FIG. 1).

[0078] At 202, information related to an upcoming vehicle path is acquired, including a range of the vehicle path. The vehicle path may represent an entire journey for the vehicle

100, from a starting location to a destination location. The information related to the upcoming vehicle path may comprise path topography such as a friction profile, a curvature, and/or a vertical profile of the vehicle path.

[0079] At 204, a vehicle motion profile is determined as a solution to an optimisation problem that describes a total cost for a vehicle 100 to follow the vehicle path. The optimisation problem includes a first cost associated with energy usage of the vehicle 100 and a second cost associated with tyre wear of the vehicle 100. The first cost represents one or more of CO2 emissions and/or electrical energy usage. The second cost represents particulate emissions from a tyre, tyre tread depth reduction, and/or a cost associated with tyre maintenance. The range of the vehicle path may be implemented as a constraint of the optimisation problem.

[0080] The optimisation problem may be an OCP. The OCP may be an NOCP, although it is also possible to formulate a linear OCP that provides a solution of sufficient accuracy. The solution is found to the optimisation problem such that the total cost, including the cost associated with energy usage and the cost associated with tyre wear, is below a threshold, for example minimised. Therefore, the cost function is the cost of energy usage and tyre wear together, while within the NOCP PCM, optimal decisions are found in a way to minimise both energy consumption and tyre wear at the same time with optimal distribution of requested power between electric motors of different axles and service brake of different axles, while maintaining the maximum range.

[0081] The vehicle motion profile comprises a target tyre force trajectory and/or a target wheel slip trajectory for at least two wheels to be followed along the upcoming vehicle path. The vehicle motion profile may comprise a target vehicle unit acceleration and/or a target vehicle speed to be followed along the upcoming vehicle path. Both the target force trajectory and the acceleration profile may be generated under slip constraints, which means that the vehicle can be controlled to behave according to the target force trajectory and/or the acceleration profile without experiencing wheel slip in excess of the configured slip limits.

The vehicle motion profile may comprise a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle.

[0082] It is important to note that energy efficiency does not mean cost efficiency. There can be situations in which an energy efficient power distribution increases the with tyre wear cost. In that case, inclusion of with wear in the cost function helps to reduce the total cost of operations. However, reducing the cost rather than energy may reduce the range. Therefore, the cost optimisation can be performed in a predictive manner with a set requested range, reducing the total cost by including the tyre wear in the cost function without conflicting with the range requirement. The method **200** prioritises the electric machines and driven axles not only by energy efficiency but by the total cost of operations, especially when there is more than one driven axle in the vehicle. Taking an entire journey into account means that the requested range is properly met, which may not be the case where similar considerations are considered on an instantaneous or short-term basis.

[0083] At **206**, the vehicle motion profile may be provided to a control system configured to determine one or more control inputs for the vehicle **100**. The control inputs may comprise a force request, a torque request, and/or a longitudinal slip request for the vehicle **100**. In particular, the control inputs may be applied to one or more driven axles of the vehicle **100**. Tyre wear caused by axial torque and/or longitudinal slip is especially higher on driven axles of a vehicle compared to other axles. The reason is that the driven axle tyres are under more stress during acceleration and deceleration or retardation compared to other axles. For the same reason, and also for the fact that the response time of electric-driveline is faster than that of diesel vehicles, the driven axles of battery electric vehicles (BEVs) experience higher tyre wear.

[0084] FIG. 3 is a schematic diagram of a computer system **300** for implementing examples disclosed herein. The computer system **300** is adapted to execute instructions from a computer-readable medium to perform these and/or any of the functions or processing described herein. The computer system **300** may be connected (e.g., networked) to other machines in a LAN, an intranet, an extranet, or the Internet. While only a single device is illustrated, the computer system **300** may include any collection of devices that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. Accordingly, any reference in the disclosure and/or claims to a computer system, computing system, computer device, computing device, control system, control unit, electronic control unit (ECU), processor device, processing circuitry, etc., includes reference to one or more such devices to individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein. For example, control system may include a single control unit or a plurality of control units connected or otherwise communicatively coupled to each other, such that any performed function may be distributed between the control units as desired. Further, such devices may communicate with each other or other devices by various system architectures, such as directly or via a Controller Area Network (CAN) bus, etc.

[0085] The computer system **300** may comprise at least one computing device or electronic device capable of including firmware, hardware, and/or executing software

instructions to implement the functionality described herein. The computer system **300** may include processing circuitry **302** (e.g., processing circuitry including one or more processor devices or control units), a memory **304**, and a system bus **306**. The computer system **300** may include at least one computing device having the processing circuitry **302**. The system bus **306** provides an interface for system components including, but not limited to, the memory **304** and the processing circuitry **302**. The processing circuitry **302** may include any number of hardware components for conducting data or signal processing or for executing computer code stored in memory **304**. The processing circuitry **302** may, for example, include a general-purpose processor, an application specific processor, a Digital Signal Processor (DSP), an Application Specific Integrated Circuit (ASIC), a Field Programmable Gate Array (FPGA), a circuit containing processing components, a group of distributed processing components, a group of distributed computers configured for processing, or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. The processing circuitry **302** may further include computer executable code that controls operation of the programmable device.

[0086] The system bus **306** may be any of several types of bus structures that may further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and/or a local bus using any of a variety of bus architectures. The memory **304** may be one or more devices for storing data and/or computer code for completing or facilitating methods described herein. The memory **304** may include database components, object code components, script components, or other types of information structure for supporting the various activities herein. Any distributed or local memory device may be utilized with the systems and methods of this description. The memory **304** may be communicably connected to the processing circuitry **302** (e.g., via a circuit or any other wired, wireless, or network connection) and may include computer code for executing one or more processes described herein. The memory **304** may include non-volatile memory **308** (e.g., read-only memory (ROM), crasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), etc.), and volatile memory **310** (e.g., random-access memory (RAM)), or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a computer or other machine with processing circuitry **302**. A basic input/output system (BIOS) **312** may be stored in the non-volatile memory **308** and can include the basic routines that help to transfer information between elements within the computer system **300**.

[0087] The computer system **300** may further include or be coupled to a non-transitory computer-readable storage medium such as the storage device **314**, which may comprise, for example, an internal or external hard disk drive (HDD) (e.g., enhanced integrated drive electronics (EIDE) or serial advanced technology attachment (SATA)), HDD (e.g., EIDE or SATA) for storage, flash memory, or the like. The storage device **314** and other drives associated with computer-readable media and computer-usable media may provide non-volatile storage of data, data structures, computer-executable instructions, and the like.

[0088] Computer-code which is hard or soft coded may be provided in the form of one or more modules. The module(s) can be implemented as software and/or hard-coded in circuitry to implement the functionality described herein in whole or in part. The modules may be stored in the storage device **314** and/or in the volatile memory **310**, which may include an operating system **316** and/or one or more program modules **318**. All or a portion of the examples disclosed herein may be implemented as a computer program **320** stored on a transitory or non-transitory computer-usable or computer-readable storage medium (e.g., single medium or multiple media), such as the storage device **314**, which includes complex programming instructions (e.g., complex computer-readable program code) to cause the processing circuitry **302** to carry out actions described herein. Thus, the computer-readable program code of the computer program **320** can comprise software instructions for implementing the functionality of the examples described herein when executed by the processing circuitry **302**. In some examples, the storage device **314** may be a computer program product (e.g., readable storage medium) storing the computer program **320** thereon, where at least a portion of a computer program **320** may be loadable (e.g., into a processor) for implementing the functionality of the examples described herein when executed by the processing circuitry **302**. The processing circuitry **302** may serve as a controller or control system for the computer system **300** that is to implement the functionality described herein.

[0089] The computer system **300** may include an input device interface **322** configured to receive input and selections to be communicated to the computer system **300** when executing instructions, such as from a keyboard, mouse, touch-sensitive surface, etc. Such input devices may be connected to the processing circuitry **302** through the input device interface **322** coupled to the system bus **306** but can be connected through other interfaces, such as a parallel port, an Institute of Electrical and Electronic Engineers (IEEE) 1394 serial port, a Universal Serial Bus (USB) port, an IR interface, and the like. The computer system **300** may include an output device interface **324** configured to forward output, such as to a display, a video display unit (e.g., a liquid crystal display (LCD) or a cathode ray tube (CRT)). The computer system **300** may include a communications interface **326** suitable for communicating with a network as appropriate or desired.

[0090] The operational actions described in any of the exemplary aspects herein are described to provide examples and discussion. The actions may be performed by hardware components, may be embodied in machine-executable instructions to cause a processor to perform the actions, or may be performed by a combination of hardware and software. Although a specific order of method actions may be shown or described, the order of the actions may differ. In addition, two or more actions may be performed concurrently or with partial concurrence.

[0091] According to certain examples, there is also disclosed:

[0092] Example 1: A computer system (**140, 160, 300**) for determining a vehicle motion profile, the computer system (**140, 160, 300**) comprising processing circuitry (**150, 170, 302**) configured to: acquire information related to an upcoming vehicle path, including a range of the vehicle path; and determine a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle

(**100**) to follow the vehicle path including a first cost associated with energy usage of the vehicle (**100**) and a second cost associated with tyre wear of the vehicle (**100**); wherein the total cost associated with the solution to the optimisation problem is below a threshold; and the determined vehicle motion profile meets the range of the vehicle path.

[0093] Example 2: The computer system (**140, 160, 300**) of example 1, wherein the vehicle path represents an entire journey for the vehicle (**100**), from a starting location to a destination location.

[0094] Example 3: The computer system (**140, 160, 300**) of example 1 or 2, wherein the information related to the upcoming vehicle path comprises a friction profile, a curvature, and/or a vertical profile of the vehicle path.

[0095] Example 4: The computer system (**140, 160, 300**) of any preceding example, wherein the processing circuitry (**150, 170, 302**) is configured to implement the range of the vehicle path as a constraint of the optimisation problem.

[0096] Example 5: The computer system (**140, 160, 300**) of any preceding example, wherein the optimisation problem is a non-linear optimal control problem, NOCP.

[0097] Example 6: The computer system (**140, 160, 300**) of any preceding example, wherein the first cost represents CO₂ emissions and/or electrical energy usage.

[0098] Example 7: The computer system (**140, 160, 300**) of any preceding example, wherein the second cost represents particulate emissions from a tyre, friction loss from a tyre, and/or a cost associated with tyre maintenance.

[0099] Example 8: The computer system (**140, 160, 300**) of example 7, wherein the processing circuitry (**150, 170, 302**) is configured to determine a cost associated with particulate emissions and/or friction loss from a tyre as a function of a longitudinal slip associated with the one or more wheels (**100**).

[0100] Example 9: The computer system (**140, 160, 300**) of any preceding example, wherein the vehicle motion profile comprises one or more of a speed of the vehicle (**100**), a trajectory of the vehicle (**100**), and a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle (**100**).

[0101] Example 10: The computer system (**140, 160, 300**) of any preceding example, wherein the processing circuitry (**150, 170, 302**) is further configured to provide the vehicle motion profile to a control system (**150, 170**) configured to determine one or more control inputs for the vehicle (**100**).

[0102] Example 11: The computer system (**140, 160, 300**) of example 10, wherein the one or more control inputs relate to one or more driven axles of the vehicle (**100**).

[0103] Example 12: A vehicle (**100**) comprising the computer system (**140, 160, 300**) of any preceding example.

[0104] Example 13: A computer-implemented method (**200**) for determining a vehicle motion profile, the method (**200**) comprising: acquiring (**202**), by processing circuitry (**150, 170, 302**) of a computer system (**140, 160, 300**), information related to an upcoming vehicle path, including a range of the vehicle path; and determining (**204**), by the processing circuitry (**150, 170, 302**), a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle (**100**) to follow the vehicle path including a first cost associated with energy usage of the vehicle (**100**) and a second cost associated with tyre wear of the vehicle (**100**); wherein the total cost associated with the solution to

the optimisation problem is below a threshold; and the determined vehicle motion profile meets the range of the vehicle path.

[0105] Example 14: The computer-implemented method (200) of example 13, wherein the vehicle path represents an entire journey for the vehicle (100), from a starting location to a destination location.

[0106] Example 15: The computer-implemented method (200) of example 13 or 14, wherein the information related to the upcoming vehicle path comprises a friction profile, a curvature, and/or a vertical profile of the vehicle path.

[0107] Example 16: The computer-implemented method (200) of any of examples 13 to 15, comprising implement, by the processing circuitry (150, 170, 302), the range of the vehicle path as a constraint of the optimisation problem.

[0108] Example 17: The computer-implemented method (200) of any of examples 13 to 16, wherein the optimisation problem is a non-linear optimal control problem, NOCP.

[0109] Example 18: The computer-implemented method (200) of any of examples 13 to 17, wherein the first cost represents CO2 emissions and/or electrical energy usage.

[0110] Example 19: The computer-implemented method (200) of any of examples 13 to 18, wherein the second cost represents particulate emissions from a tyre, friction loss from a tyre, and/or a cost associated with tyre maintenance.

[0111] Example 20: The computer-implemented method (200) of example 19, comprising determining, by the processing circuitry (150, 170, 302), a cost associated with particulate emissions and/or friction loss from a tyre as a function of a longitudinal slip associated with the one or more wheels (100).

[0112] Example 21: The computer-implemented method (200) of any of examples 13 to 20, wherein the vehicle motion profile comprises one or more of a speed of the vehicle (100), a trajectory of the vehicle (100), and a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle (100).

[0113] Example 22: The computer-implemented method (200) of any of examples 13 to 21, further comprising providing, by the processing circuitry (150, 170, 302), the vehicle motion profile to a control system (150, 170) configured to determine one or more control inputs for the vehicle (100).

[0114] Example 23: The computer-implemented method (200) of example 22, wherein the one or more control inputs relate to one or more driven axles of the vehicle (100).

[0115] Example 24: A computer program product comprising program code for performing, when executed by processing circuitry (150, 170, 302), the computer-implemented method (200) of any of examples 13 to 23.

[0116] Example 25: A non-transitory computer-readable storage medium comprising instructions, which when executed by processing circuitry (150, 170, 302), cause the processing circuitry to perform the computer-implemented method (200) of any of examples 13 to 23.

[0117] Terminology used herein is for the purpose of describing particular aspects only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. It will be further understood that the terms “comprises,” “comprising,” “includes,” and/or “including” when used herein

specify the presence of stated features, integers, actions, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, actions, steps, operations, elements, components, and/or groups thereof.

[0118] It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first element could be termed a second element, and, similarly, a second element could be termed a first element without departing from the scope of the present disclosure.

[0119] Relative terms such as “below” or “above” or “upper” or “lower” or “horizontal” or “vertical” may be used herein to describe a relationship of one element to another element as illustrated in the Figures. It will be understood that these terms and those discussed above are intended to encompass different orientations of the device in addition to the orientation depicted in the Figures. It will be understood that when an element is referred to as being “connected” or “coupled” to another element, it can be directly connected or coupled to the other element, or intervening elements may be present. In contrast, when an element is referred to as being “directly connected” or “directly coupled” to another element, there are no intervening elements present.

[0120] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which this disclosure belongs. It will be further understood that terms used herein should be interpreted as having a meaning consistent with their meaning in the context of this specification and the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0121] It is to be understood that the present disclosure is not limited to the aspects described above and illustrated in the drawings; rather, the skilled person will recognize that many changes and modifications may be made within the scope of the present disclosure and appended claims. In the drawings and specification, there have been disclosed aspects for purposes of illustration only and not for purposes of limitation, the scope of the disclosure being set forth in the following claims.

What is claimed is:

1. A computer system for determining a vehicle motion profile, the computer system comprising processing circuitry configured to:

acquire information related to an upcoming vehicle path, including a range of the vehicle path; and

determine a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle to follow the vehicle path including a first cost associated with energy usage of the vehicle and a second cost associated with tyre wear of the vehicle;

wherein the total cost associated with the solution to the optimisation problem is below a threshold; and

the determined vehicle motion profile meets the range of the vehicle path.

2. The computer system of claim 1, wherein the vehicle path represents an entire journey for the vehicle, from a starting location to a destination location.

3. The computer system of claim 1, wherein the information related to the upcoming vehicle path comprises a friction profile, a curvature, and/or a vertical profile of the vehicle path.

4. The computer system of claim 1, wherein the processing circuitry is configured to implement the range of the vehicle path as a constraint of the optimisation problem.

5. The computer system of claim 1, wherein the optimisation problem is a non-linear optimal control problem, NOCP.

6. The computer system of claim 1, wherein the first cost represents CO2 emissions and/or electrical energy usage.

7. The computer system of claim 1, wherein the second cost represents particulate emissions from a tyre, friction loss from a tyre, and/or a cost associated with tyre maintenance.

8. The computer system of claim 7, wherein the processing circuitry is configured to determine a cost associated with particulate emissions and/or friction loss from a tyre as a function of a longitudinal slip associated with the one or more wheels.

9. The computer system of claim 1, wherein the vehicle motion profile comprises one or more of a speed of the vehicle, a trajectory of the vehicle, and a power split between driven axles, electric machines, and/or an internal combustion engine of the vehicle.

10. The computer system of claim 1, wherein the processing circuitry is further configured to provide the vehicle motion profile to a control system configured to determine one or more control inputs for the vehicle.

11. The computer system of claim 10, wherein the one or more control inputs relate to one or more driven axles of the vehicle.

12. A vehicle comprising the computer system of claim 1.

13. A computer-implemented method for determining a vehicle motion profile, the method comprising:

acquiring, by processing circuitry of a computer system, information related to an upcoming vehicle path, including a range of the vehicle path; and

determining, by the processing circuitry, a vehicle motion profile as a solution to an optimisation problem that describes a total cost for a vehicle to follow the vehicle path including a first cost associated with energy usage of the vehicle and a second cost associated with tyre wear of the vehicle;

wherein the total cost associated with the solution to the optimisation problem is below a threshold; and

the determined vehicle motion profile meets the range of the vehicle path.

14. The computer-implemented method of claim 13, wherein the vehicle path represents an entire journey for the vehicle, from a starting location to a destination location.

15. The computer-implemented method of claim 13, comprising implement, by the processing circuitry, the range of the vehicle path as a constraint of the optimisation problem.

16. The computer-implemented method of claim 13, wherein the optimisation problem is a non-linear optimal control problem, NOCP.

17. The computer-implemented method of claim 13, wherein the second cost represents particulate emissions from a tyre, friction loss from a tyre, and/or a cost associated with tyre maintenance.

18. The computer-implemented method of claim 17, comprising determining, by the processing circuitry, a cost associated with particulate emissions and/or friction loss from a tyre as a function of a longitudinal slip associated with the one or more wheels.

19. A computer program product comprising program code for performing, when executed by processing circuitry, the computer-implemented method of claim 13.

20. A non-transitory computer-readable storage medium comprising instructions, which when executed by processing circuitry, cause the processing circuitry to perform the computer-implemented method of claim 13.

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